C201/C255/I211/A506 Homework #2 Due 2/12/26

This program inputs a manuscript in text form from standard input device and analyzes the lengths of all the words encountered. Only alphanumeric characters, apostrophes and hyphens within a word contribute to the word length. Basically you need to write a program that reads words from a standard input, either from the keyboard or from a file via DOS redirection and then outputs a table showing the frequencies of words of various lengths. The output could either be to screen or redirected to a file. Words longer than 15 characters are treated as 15 character word. Example output follows: This is output when run program with file WORDS.4:

Word Length	Frequency
1	5
2	7
3	15
4	12
5	8
6	6
7	4
8	4
9	2
10	0
11	1
12	0
13	0
14	0
15	0

Average word length: 4.328125

Notes:

A) The average word length is from:

(Total length of all words combined) / (Number of words)

B) For test data come up with four of your own plus test my five files called WORDS.1, WORDS.2, WORDS.3, WORDS.4, and WORDS.5 These files are found on in Canvas under the Homework and Labs link. You will need to copy these files to your disk or location were your program will run from. All of the files have text in them except for WORDS.3, this file is empty. **Make sure to hand in your data files that you test as well. Don't turn in my files.**

C) To get your program to read from a file instead of keyboard you'll have to run the program from the Command prompt and use redirection. Refer to the handout on *"How to run Java programs from the command prompt"* and "*I/O Redirection*". Capturing the screen output can be done several ways. In Windows, in output window, right click, then select Mark, then select text, then press Enter, where it is placed on clipboard. This can then be placed in Note Pad to print out. Could also redirect output to another file.

D) When turning in homework refer to handouts, *Rules for programming assignments* and *Turning in Programming Assignments*. Make sure to comment well, and tell what is considered a word.

E) Think carefully on algorithm design and what determines a word. We will discuss this in more detail in class.

How to submit your homework's (For online students only)

All homework's are due by the posted date on the top of assignment and by midnight. If turned in one class period late then -3 points and -6 points if turned in two class periods late by midnight, nothing will be accepted after that time. Everything will be submitted via email, do NOT use canvas or canvas messages to send me anything. Use your school or personal email only and send to my school email or personal email given on the syllabus.

Send the completed homework as an attachment with one and only one pdf file. Make sure you have your name on the assignment as well.

IMPORTANT: Make sure to submit one pdf file that contains all the programs and or algorithms. Each homework problem will have its own program file with output that will need to be saved as a pdf file. All the separate pdf files will then need to be combined into one pdf file to send to me. Make sure the combined pdf files are in order as given in homework page.

If you have a handwritten algorithm to submit, then there are two ways listed below to save file as pdf.

1) You can scan the document on a printer and save as a pdf.
2) You can download an app on your phone that allows you to take picture of document and it will then save it as a pdf file.

Combining pdf files several ways, choose a method you know how, or try on listed below.

1) Go to https://iuware.iu.edu/ and download the "Creative Cloud" from adobe, then can install "Acrobat Pro" for dealing with pdf files. (I called OIT and had them help with this).
2) Can try LightPDF, https://lightpdf.com/, I have not used this but other students in past have used it.

Even if you are unable to complete the assignment, turn what you have done in. Something is worth some points, nothing is a zero.